

1. UniversityMember

```
public abstract class UniversityMember {  
    protected int MemberID;  
    protected String name;  
    public UniversityMember (int MemberID, String name)  
    {  
        this.MemberID = MemberID;  
        this.name = name;  
    }  
    public void displayBasicInfo () {  
        ID: IT24003  
        System.out.println ("ID: " + MemberID);  
        System.out.println ("Name: " + name);  
    }  
    public abstract void PerformDuty();  
}
```

2. Employee

```
public class Employee extends UniversityMember {  
    protected double salary;  
    public Employee(int MemberID, String name, String  
        double salary) {  
        super(MemberID, name);  
        this.salary = salary;  
    }  
    public void ShowSalary() {  
        System.out.println("Salary: " + salary);  
    }  
}
```

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@Override

```
public void PerformDuty() {  
    System.out.println(name + "Perform employee duty");  
}
```

3. Academic Staff:

```
public class AcademicStaff extends Employee {  
    protected String department;  
    public AcademicStaff (int MemberID, String name,  
        double salary, String department) {  
        super(MemberID, name, salary);  
        this.department = department;  
    }  
    public void ShowDepartment () {
```

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```
        System.out.println("Department: " + department);  
    }  
    @Override  
    public void PerformDuty () {  
        System.out.println(name + " teaches students");  
    }  
}
```


4. Professor

public class Professor extends AcademicStaff {

private String ResearchArea;

private int publications;

public Professor(String name, int MemberID, double salary, String department, String ResearchArea, int publications) {

super(MemberID, name, salary, department);

this.ResearchArea = ResearchArea;

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this.publications = publications;

public void ShowResearchProfile() {

System.out.println("ResearchArea: " + ResearchArea);

System.out.println("Publications: " + publications);

@Override

public void PerformDuty() {

System.out.println(name + "Conducts research and teaches advanced courses");

Administrative Officer

public class AdministrativeOfficer extends Employee {
private String OfficeName;

public AdministrativeOfficer(int MemberID, String
name, double salary, String department, String
Research Area, int publications, String Office
Name)

super(MemberID, name, salary, department,
Research Area, publications);

this.OfficeName = OfficeName;

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public void ShowOffice() {

System.out.println("Office " + OfficeName)

}

@Override

public void PerformDuty() {

System.out.println(name + " manages university
administrative activities");

}

6. Main

```
import java.util. ArrayList;
```

```
public class Main {
```

```
    public static void Main(String[] args) {
```

```
        ArrayList < University Member > members = new  
            ArrayList < > ();
```

```
        members.add(new Professor ( 100006, "Dr. XYZ",  
            1,30,000, "CSE", "CyberSecurity", 45 ));
```

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```
        members.add(new AdministrativeOfficer ( 50001,  
            "Mr. Koriu", 60,000, "Registrar Office" ));
```

```
        System.out.println ("=== Professors Details ===");
```

```
        for ( University Member member : members ) {  
            if ( member instanceof Professor ) {
```

```
                Professor p = ( Professor ) member;
```

```
                p.DisplayBasicInfo ();
```

```
                p.ShowSalary ();
```

P. Show Department() ;

P. Show Research Profile() ;

P. Perform Duty() ;

System.out.println("==== Administrative Officer
Details =====");

for (University Member member : members) {
 it(member instanceof Administrative Officer) ;

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Administrative Officer ao = (Administrative
Officer) member ;

ao.Display Basic Info() ;

ao.Show Salary() ;

ao.Show Office() ;

ao.Perform Duty() ;